

A Simple SNAKE & LADDER Game Implementation using PYTHON

```
import random

# Global variables
dice = 0
key = ""
choice = ""
position = 0
newposition = 0
count = 0

#Commentary while playing
def check_position(position):
    global newposition
    if position == 4:
        newposition = 14
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 9:
        newposition = 31
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 17:
        newposition = 7
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 21:
        newposition = 42
        print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
    elif position == 46:
        newposition = 13
        print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
    elif position == 28:
        newposition = 84
```

```
    print(f"\nWow! You landed on a big ladder. Now you are on {newposition}.")
elif position == 62:
    newposition = 19
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
elif position == 37:
    newposition = 61
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
elif position == 54:
    newposition = 25
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
elif position == 51:
    newposition = 67
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
elif position == 87:
    newposition = 36
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
elif position == 72:
    newposition = 91
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
elif position == 64:
    newposition = 44
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
elif position == 80:
    newposition = 23
    print(f"\nOh! You landed on a big snake. Now you are on {newposition}.")
elif position == 38:
    newposition = 71
    print(f"\nWell done! You landed on a ladder. Now you are on {newposition}.")
elif position == 95:
    newposition = 75
    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
```

```

elif position == 98:
    newposition = 79

    print(f"\nBad luck! You landed on a snake. Now you are on {newposition}.")
else:
    newposition = position

return newposition

```

```
def main():
```

```
    global dice, position, newposition, count, key, choice
```

```
    # Initial things
```

```
    print("Would you like to Play the game???(click Y or y if you want)")
```

```
    choice = input().strip().lower()
```

```
    if choice == 'y' or choice=='Y':
```

```
        print(" ### Welcome to Snake and Ladders Game !!!!! | Rules Are:: |")
```

```
        print("| 100 99 98 97 96 95 94 93 92 91 | | 1=START 4=Ladder to 14 |")
```

```
        print("| 81 82 83 84 85 86 87 88 89 90 | | 9=Ladder to 31 17=Snake to 7 |")
```

```
        print("| 80 79 78 77 76 75 74 73 72 71 | | 21=Ladder to 42 46=Snake to 13 |")
```

```
        print("| 61 62 63 64 65 66 67 68 69 70 | | 28=Ladder to 84 62=Snake to 19 |")
```

```
        print("| 60 59 58 57 56 55 54 53 52 51 | | 37=Ladder to 61 54=Snake to 25 |")
```

```
        print("| 41 42 43 44 45 46 47 48 49 50 | | 51=Ladder to 67 87=Snake to 36 |")
```

```
        print("| 40 39 38 37 36 35 34 33 32 31 | | 72=Ladder to 91 64=Snake to 44 |")
```

```
        print("| 21 22 23 24 25 26 27 28 29 30 | | 80=Snake to 23 95=Snake to 75 |")
```

```
        print("| 20 19 18 17 16 15 14 13 12 11 | | 38=Ladder to 71 98=Snake to 79 |")
```

```
        print("| 1 2 3 4 5 6 7 8 9 10 | | 100=END |")
```

```
        print("**WINNER WILL BE THE ONE WHO GETS TO 100 FIRST.THAT IS WITH MINIMUM NUMBER  
OF DICE THROWS..\n")
```

```
# Start the game loop
```

```
while position < 100:
```

```
    input("\nPress any key to roll the dice\n")
```

```
    print("Player rolled a dice.")
```

```
    dice = random.randint(1, 6)
```

```
    position += dice
```

```
    print(f"You rolled a {dice}. You are now on square {position}.")
```

```
    count += 1
```

```
# If you roll a 6, you get another chance to roll
```

```
if dice == 6:
```

```
    print("\n6 came up! You get another chance to roll.")
```

```
    continue
```

```
# Check the position for ladders or snakes
```

```
newposition = check_position(position)
```

```
position = newposition
```

```
print(f"\nCONGRATULATIONS, you have won the game in {count} throws!")
```

```
else:
```

```
    print("\nOK, Go to Another Game!")
```

```
if __name__ == "__main__":
```

```
    main()
```

Key Points to Note:

Input/Output: In C, there are `scanf` and `printf` for input and output, respectively. In Python, `input()` is used for getting user input and `print()` for displaying messages.

Random Dice Rolls: In C, `rand()` is used to generate random numbers, while in Python, `random.randint(1, 6)` is the equivalent.

Game Loop: The game loop continues while the player's position is less than 100. The player's position updates based on the dice roll and the ladder or snake positions.

Ladders and Snakes: The `check_position()` function in Python handles updating the player's position if they land on a ladder or snake.

How to Play:

The player is prompted whether they want to play.

The player rolls a dice by pressing any key. The dice roll determines the number of steps they move forward.

If they land on a snake, they move backwards; if they land on a ladder, they move forwards.

The goal is to reach position 100 as quickly as possible.